

your foundation!

Ask anybody about the computer he or she owns and you get to hear descriptions that go something like “a Pentium III 750 with 128 MB of RAM and a 30 GB hard drive”. While this gives you a fair idea of the system, it misses a vital piece of information—the motherboard. What makes this component so important is the fact that all the other components plug into and communicate with each other via the motherboard. The motherboard or mainboard acts as the central (and largest) part of your computer. All components of the machine—video cards, soundcards, hard drives, CD-ROM drives and even external peripherals like printers—are eventually connected to the motherboard.

When you want to plug in a new piece of hardware in your machine, guess what's the deciding factor? The type of motherboard. This determines which CPUs will and will not work with your system. Thus, the task of buying a motherboard shouldn't be taken lightly.

Look at it in this way. Either you can save money by buying a cheap, unbranded motherboard and suffer sluggish perform-

ance and lose out on upgradability, or choose wisely and buy a board which performs well, is compatible with the next generation CPUs, and saves you the trouble and cost of buying another motherboard in a year or two.

What is a chipset?

The motherboard is essentially built around a chipset that supports a specific class of processors and memory. The main system chipset comprises the logic circuits that basically contain a blueprint of how the processor, memory and other integrated components on the motherboard are meant to communicate. It acts as a hub, controlling the transfer of data between the processor, its cache, system buses and various peripherals.

The timely transfer of data without errors is probably the most important function of a system chipset and its proficiency in performing this task dictates how well the system performs as a whole. Apart from the main system chipset which is commonly referred to as the North Bridge, motherboards also have a separate South Bridge that integrates, manages and controls all 'external' IDE, ISA, PCI, USB, etc devices.

However, the North Bridge or the main system chipset is the one that we shall refer to as it directly impacts performance.

Is the chipset important?

Chipset design is a complicated task that has been mastered only by a few manufacturers. Processor manufacturers such as Intel are ideally positioned to design chipsets that suit their needs. Intel designs chipsets that work with its own processors—that's why you find few inherent problems with Intel chipset-based motherboards.

AMD, on the other hand, has preferred to let third-party manufacturers like VIA, SiS and ALi produce the bulk of the chipsets supporting its processors. VIA, especially, has been very successful at squeezing out the maximum performance from AMD's processors.

The performance numbers derived from our tests indicate a clear pattern which separates the good chipsets from the ordinary ones. For the Pentium 4, we discovered that the Intel 850 series-based motherboards, coupled with RDRAM, outperformed every other motherboard by a fair degree.

Intel's stop-gap SDRAM solution, the 845 series-based motherboards performed poorly when compared to the 845D-based solutions using DDR SDRAM. The VIA P4M266 and the P4X266 chipsets notch up some really interesting scores, at times even beating Intel's own SDRAM chipset.

We would recommend the Intel 845 for SDRAM, Intel 845D for DDR SDRAM and the Intel 850 series for RDRAM.

For the AMDs, the top-of-the-line chipset is still the VIA KT266A, though the KT333 should technically outperform its older brother when clubbed with the newer DDR333 memory. Another interesting observation is that nVidia's first foray into chipsets has proved to be a decent performer. The nForce 420-D-